
IS BIOTECHNOLOGY A THREAT TO THE ENVIRONMENT?

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The commercial release of genetically modified organisms, specifically, Golden Rice and Bt (*Bacillus thuringiensis*) eggplant, is no more, at least for now.

On April 18, 2023, the Supreme Court issued a *Writ of Kalikasan* in the case of *Magsasaka at Siyentipiko Para sa Pag-Unlad ng Agrikultura, et al. v. Secretary of the Department of Agriculture, et. al.*¹

In a briefer released on April 19, 2023, the Court required the respondents Secretary of the Department of Agriculture, the Secretary of the Department of Environment and Natural Resources, the Secretary of the Department of Health, the Director of the Bureau of Plant Industry of the Department of Agriculture, the Philippine Rice Research Institute, and University of the Philippines – Los Baños (UPLB) to file a verified return within 10 days from service.²

It all started when

The Department of Agriculture (DA) issued Department of Agriculture Administrative Order No. 08, series of 2002 (DAO-08-2002) providing rules and regulations for the importation and release into the environment of plants and plant products derived from the use of modern biotechnology.

DAO-08-2002 covers the importation or release into the environment of: (1) any plant which has been altered or produced through the use of modern biotechnology if the donor organism, host organism, or vector or vector agent belongs to the genera or taxa classified by the Bureau of Plant Industry (BPI) as meeting the definition of plant pest or is a medium for the introduction of noxious weeds; or (2) any plant or plant product altered through the use of modern biotechnology which may pose significant risks to human health and the environment based on available scientific and technical information.

Farmers, Non-Governmental Organizations (NGOs), organic traders and other sectors of society expressed their opposition, noting the threats Genetically

¹ *Magsasaka at Siyentipiko Para sa Pag-Unlad ng Agrikultura, et al. v. Secretary of the Department of Agriculture, et. al.* G.R. 263595, (April 18, 2023) (Phil). The Supreme Court Public Information Office will upload a copy of the ruling once it receives a copy of the same from the *En Banc* Office of the Clerk of Court.

² SC Issues Writ of Kalikasan on Genetically Modified Rice and Eggplant Products (April 19, 2023), <https://sc.judiciary.gov.ph/sc-issues-writ-of-kalikasan-on-genetically-modified-rice-and-eggplant-products/> (Portal of the Philippine Judiciary).

Modified Organisms (GMOs) pose to human beings and the environment, as well as the possible contamination that could affect the Philippine organic industry.³

Unconstitutional

In *Int'l. Service for the Acquisition of Agri-Biotech Applications, Inc. vs. Greenpeace Southeast Asia (Phils.), et al.*,⁴ the Supreme Court declared DAO-08-2002⁵ as null and void. It ruled that “the failure of DAO 08-2002 to accommodate the NBF means that the Department of Agriculture lacks mechanisms to mandate applicants to comply with international biosafety protocols.”

Precautionary Principle

Notably, the Supreme Court discussed the Precautionary Principle.

It stated that “three conditions must be present: (a) settings in which the risks of harm are uncertain; (b) settings in which harm might be irreversible and what is lost is irreplaceable; and (c) settings in which the harm that might result would be serious.”

The precautionary principle bridges the gap in cases where scientific certainty in factual findings cannot be achieved. By applying the precautionary principle, the court may construe a set of facts as warranting either judicial action or inaction, with the goal of preserving and protecting the environment. This may be further evinced from the second paragraph where bias is created in favor of the constitutional right of the people to a balanced and healthful ecology. In effect, the precautionary principle shifts the burden of evidence of harm away from those likely to suffer harm and onto those desiring to change the status quo. An application of the precautionary principle to the rules on evidence will enable courts to tackle future environmental problems before ironclad scientific consensus emerges.

³ BPI's conflict of interest in GMO products, (February 13, 2012) <https://opinion.inquirer.net/22993/bpi%E2%80%99s-conflict-of-interest-in-gmo-products> (Philippine Daily Inquirer, a daily newspaper in the Philippines).

⁴ *Int'l. Service for the Acquisition of Agri-Biotech Applications, Inc. v. Greenpeace Southeast Asia (Phils.), et al.*, G.R. No. 209271, (December 08, 2015) (Phil).

⁵ Department of Agriculture Administrative Order No. 8 Series of 2002 (April 3, 2002) <http://spsissuances.da.gov.ph/attachments/article/298/DA%20Administrative%20Order%20No.%208%20S2002.pdf> (Sanitary and Phytosanitary Regulations and related Issuances, Department of Agriculture).

As applied

Assessing the evidence on record, as well as the current state of GMO research worldwide, the Court finds all the three conditions present.

Eggplants (*talong*) are a staple vegetable in the country and grown by small-scale farmers, majority of whom are poor and marginalized. While the goal of increasing crop yields to raise farm incomes is laudable, independent scientific studies revealed uncertainties due to unfulfilled economic benefits from *Bt* crops and plants, adverse effects on the environment associated with use of GE technology in agriculture, and serious health hazards from consumption of GM foods. For a biodiversity-rich country like the Philippines, the natural and unforeseen consequences of contamination and genetic pollution would be disastrous and irreversible.

Alongside the aforesaid uncertainties, the non-implementation of the NBF in the crucial stages of risk assessment and public consultation, including the determination of the applicability of the EIS requirements to GMO field testing, are compelling reasons for the application of the precautionary principle. There exists a preponderance of evidence that the release of GMOs into the environment *threatens* to damage our ecosystems and not just the field trial sites, and eventually the health of our people once the *Bt* eggplants are consumed as food. Adopting the precautionary approach, the Court rules that the principles of the NBF need to be operationalized first by the coordinated actions of the concerned departments and agencies before allowing the release into the environment of genetically modified eggplant. The more prudent course is to immediately enjoin the *Bt talong* field trials and approval for its propagation or commercialization until the said government offices shall have performed their respective mandates to implement the NBF.

The Court has found the experience of India in the *Bt brinjal* field trials - for which an indefinite moratorium was recommended by a Supreme Court-appointed committee till the government fixes regulatory and safety aspects - as relevant because majority of Filipino farmers are also small-scale farmers. Further, the precautionary approach entailed inputs from all stakeholders, including the marginalized farmers, not just the scientific community. This proceeds from the realization that acceptance of uncertainty is not only a scientific issue, but is related to public policy and involves an ethical dimension. For scientific research alone will not resolve all the problems, but participation of different stakeholders from scientists to industry, NGOs, farmers and the public will provide a needed variety of perspective foci, and knowledge.

Turn of Events

In *Int'l. Service for the Acquisition of Agri-Biotech Applications, Inc. vs. Greenpeace Southeast Asia (Phils.), et al.*,⁶ the Supreme Court overturned its Decision dated December 8, 2015 on the ground of mootness. It stated that “it was improper for the Court to resolve the merits of the case which had become moot in view of the absence of any valid exceptions to the rule on mootness, and to thereupon rule on the objections against the validity and consequently nullify DAO 08-2002 under the premises of the precautionary principle.”

And now I wonder...

What are the consequences of this jurisprudential trend?

Thoughts

I can think of two things.

First, scientists and their “inventions” are set to suffer from government regulations — at least until the government provides a clear guideline on the introduction of biotechnological innovations. This is bad because we should aim to grow with science, not resist them.

Second, the sweeping application of the precautionary principle might send a wrong message to the public. I am afraid that many will be discouraged to venture in an ambitious and probably life-changing undertaking for fear of admonishment. Do we really have to nip these activities in the bud?

My two cents

There is a need to carefully assess the possible impacts of biotechnology on the environment. It is undeniable that GMOs are still seen by the majority in a bad light. But I think that the public must give more weight to the opinions of the expert. I discourage the adoption of conservative thinking; we must adjust to the swiftly moving facts of our changing society. Thus, this subject must be evaluated with an open mind. ♦

⁶ *Int'l. Service for the Acquisition of Agri-Biotech Applications, Inc. v. Greenpeace Southeast Asia (Phils.), et al.*, G.R. No. 209271, (December 08, 2015) (Phil.).

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